Structure-Based Druggability of KRAS and PIK3CA in High-Grade Serous Ovarian Cancer

I assessed two targets from my Module 2 ovarian cancer ranking: KRAS (rank 41 of 750) and PIK3CA (rank 514 of 750). Models came from the AlphaFold Protein Structure Database (alphafold.ebi.ac.uk), protein annotation from UniProt (UniProt Consortium, 2025), and pockets were detected with fpocket.

	KRAS	PIK3CA
UniProt / Module 2 rank	P01116 / 41	P42336 / 514
Mean pLDDT / % below 50	91.5 / 2.1	92.4 / 1.1
Confident domains	1 (1-175)	1 (2-1051)
PDB (chain) / RMSD (confident Cα)	6OIM (A) / 2.19 Å	4JPS (A) / 0.90 Å
Best kept pocket score	0.612 (GDP site)	0.261 (ligand site)
Call	Possible but uncertain	Structure supports a small-molecule pocket

1. Which parts can I trust?

KRAS: The model is highly confident (77.8% of residues above 90) with no stretch of 10 or more residues below 50. The domain finder returned one confident domain (1–175), so this is the “high pLDDT + dark PAE within the region” case of the decision table, and pocket analysis is justified in the G-domain. UniProt annotates residues 166–185 as the hypervariable region, outside the GTP-binding domain, with no fixed fold. It carries palmitoylation sites (Cys-180, Lys-182, 184, 185) and a ubiquitination site (Lys-170) that regulate membrane association (UniProt Consortium, 2025), a context absent from the prediction. PAE between this tail and the rest of the protein is high, so I would not use it for pockets. The switch I (effector region, 32–40) and switch II (about 60–70) loops also show paler PAE, and switch II lines the drug pocket, so pockets there need caution.

PIK3CA: The model is highly confident (82.0% above 90; 1.1% below 50) and the finder returned a single block (2–1051, 1,003 confident residues). UniProt annotates adaptor-binding (16–105), Ras-binding (187–289), C2 (330–487), helical (517–694) and catalytic (765–1051) domains (UniProt Consortium, 2025). Pale PAE stripes fall at the inter-domain linkers (about 300–330 and 510) and at flexible kinase loops (about 870 and the activation loop, 931–957). I therefore trust pockets inside each domain, especially the kinase domain, but not pockets spanning two domains or lying in a linker. The adaptor-binding domain binds the p85α

regulatory subunit, absent from the prediction, so its placement relative to the kinase domain is the least certain.

2. How well do the predictions match experiment?

PIK3CA vs 4JPS chain A (2.20 Å) agrees closely: RMSD 0.90 Å over 968 identical, confident residues of 999 aligned. The largest deviations (residues 240, 503, 230, 415, 494, 939) are isolated residues in the Ras-binding domain, the C2-helical linker, the C2 domain and the activation loop. Residues 230 and 240 lie next to M232K and L233K, engineered mutations in the crystallized construct, which may contribute. The core fold agrees.

KRAS vs 6OIM chain A (1.65 Å) agrees less well: RMSD 2.19 Å over 158 residues, just outside the 1–2 Å "very close" range. Most of the G-domain is within 1–2 Å. The difference comes from switch I (up to about 4.5 Å), switch II (up to 6.6 Å) and residues 166–175 (5–10 Å). The crystal contains the covalent inhibitor AMG 510 (sotorasib; Canon et al., 2019), which holds switch II in a drug-bound conformation that the drug-free prediction lacks, so this is something the crystal has that the model lacks, not a prediction error. Residues 166–175 lie inside the hypervariable region and only just passed the pLDDT 70 filter. The crystal is also the G12C mutant while the model is wild type, and that residue is excluded by the identical-amino-acid filter.

3. Pocket analysis and druggability calls

Top five pockets by fpocket druggability score, with lining residues checked against the rules (reject if mean pLDDT < 70; suspect if PAE among lining residues > 10 Å):

Target	Pocket	Score	Volume (Å³)	Mean pLDDT	PAE within (Å)	Verdict
KRAS	1	0.612	1056	90.2	6.0	keep
	7	0.032	413	95.6	1.9	keep
	3	0.031	209	83.7	8.4	keep
	6	0.001	186	96.8	1.4	keep
	2	0.000	245	95.4	1.3	keep
PIK3CA	85	0.261	1295	94.6	2.5	keep
	1	0.154	984	92.6	3.4	keep
	8	0.103	367	76.7	8.6	keep
	2	0.061	745	96.1	2.1	keep
	5	0.034	522	95.0	1.8	keep

All top pockets pass the confidence filters, so the ranking rests on score and on the experimental ligand check.

PIK3CA: structure supports a small-molecule pocket. The rank 1 pocket (85) has confident lining (mean pLDDT 94.6, PAE 2.5 Å) and covers 100% of the residues within 4.5 Å of the bound inhibitor in 4JPS. The pocket is already formed in the apo model and confirmed by experiment. The modest score (0.261) fits the large, open ATP cleft of a kinase.

KRAS: possible but uncertain. The top pocket (1, score 0.612) covers 78% of the GDP site. That is the nucleotide pocket, which binds GDP/GTP very tightly, so a high score there does not give a usable drug site. The sotorasib site is covered only 38% by pocket 10 (score 0.000, rank 10, outside the top five). The structure alone therefore does not demonstrate a druggable pocket at the validated site, although experiment shows one exists.

4. Structural call versus Module 2 ranking

KRAS ranked 41 and PIK3CA 514 of 750, so the structural call reverses the order. This is consistent, because the Module 2 score reflects disease evidence (multi-omics) while the structure asks whether a drug can bind. Both targets ranked mainly on transcriptomic evidence, with low genomic scores (KRAS 0.011, PIK3CA 0.006), so the structure adds information the ranking could not provide. I would keep both. PIK3CA is structurally tractable, but its low rank means the disease evidence should be rechecked before investing. KRAS has the stronger disease rank, but a pocket-based screen on the wild-type model is not justified. It needs mutant-aware, ligand-bound structural work or a different modality.

5. Limits that apply to these targets

- **Cryptic pockets (KRAS):** The switch-II pocket forms on ligand binding and is nearly absent from the drug-free model (38% coverage, score 0.000). Molecular dynamics of the apo protein, or a protein-ligand complex prediction such as Boltz-2 on the cluster, would test whether the pocket can open.
- **Mutant pockets (KRAS):** AlphaFold predicts wild type, whereas 6OIM is KRAS G12C covalently bound to AMG 510, so the binding geometry depends on the mutated cysteine (Canon et al., 2019). Predicting the mutant sequence and comparing it with the wild-type model and the crystal would show how G12C reshapes switch II.
- **Missing pieces (PIK3CA):** PIK3CA works with p85α, which the single-chain model lacks, so the adaptor-binding domain's placement is uncertain. The 4JPS crystal contains p85α as a second chain (chain B), so comparing the model against that chain would address this directly.

6. Is the best pocket where the ligand sits?

PIK3CA: 4JPS holds one ligand, alpelisib (1LT; BindingDB IC50 1.7–1837 nM, as listed on the RCSB entry), with 19 site residues in the catalytic domain around the G-loop (771–777), catalytic loop and activation loop. UniProt describes the enzyme as using ATP, so this is the functional site (UniProt Consortium, 2025). The rank 1 kept pocket covers it fully (Furet et al., 2013).

KRAS: 6OIM holds GDP (18 site residues, 78% covered by the rank 1 pocket) in the nucleotide site and sotorasib (21 residues) in the switch-II pocket. The best kept pocket is therefore not where the drug sits. The drug site is best covered by pocket 10 (rank 10, 38% coverage, score 0.000). fpocket scores this real site low because the pocket is cryptic and shallow and shaped by the ligand, and the drug-free model places switch II differently.

Anchor check: p53

Superimposed on 192 confident Cα atoms of the DNA-binding domain (1TUP chain B, residues 96–289; Cho et al., 1994), the AlphaFold model matches with RMSD 0.68 Å (0.73 Å over all 194 common residues), better than the 1–2 Å “very close” range, although the model contains neither zinc nor DNA. Residues deviating by more than 2 Å (121, 183, 96–97, 187; maximum 4.1 Å) are loop or domain-edge residues, consistent with loop flexibility. Of the Module 3 hotspots, C176, H179, C238 and C242 are zinc ligands (2.1–2.3 Å from zinc), R248 and R273 contact DNA (3.4 and 2.5 Å), R249 is near DNA (7.2 Å), and R175, Y220, G245 and R282 do neither. Because 176, 238 and 242, which the language model found least tolerated from sequence alone, are all zinc ligands, the Module 3 scores recovered structurally critical residues without seeing the structure, while the DNA-contact and structural hotspots show that low tolerance has different causes that a sequence score cannot separate.

References

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